Rodayna Emad Mohamed Abdelkader Abdelmawla
Alexandria, Egypt
rodynaemad@gmail.com

31 October 2026

Admissions Committee
M.Sc. Data Analytics (International Master)
University of Hildesheim

Motivation Letter: M.Sc. Data Analytics (International Master)

Dear Members of the Admissions Committee,

I am writing to apply for the M.Sc. Data Analytics (International Master) at the University of Hildesheim. I graduated in July 2025 with a B.Sc. in Computer Science from Pharos University in Alexandria, Excellent with Honors (CGPA 3.88 out of 4.00, 92.13%). I want to build my career in data science, statistics and machine learning, and this master's is the next step for me.

My interest in data started in my first year, in Introduction to Data Sciences. I liked that the answer was not written in the code. It was hidden in the data, and you had to find it. In 2023 I joined WorldQuant University's online data science program, where I worked on real, messy datasets: cleaning them with pandas, exploring them with plots and heatmaps, and building my first machine learning pipelines. Around the same time I took IBM's data science courses and AWS's machine learning course. At university my best grades came in the courses closest to this field: Probability and Statistics (97), Linear Algebra (95), Data Science Methodology (94), Machine Learning (93) and Introduction to Data Sciences (93).

Two projects taught me the most. In my graduation project, BloomTech, our team built a smart agriculture system that detects plant diseases from photos, monitors soil and water with solar powered sensors, and helps farmers through a mobile app. Getting the disease detection above 90% accuracy on more than 1,000 test images showed me that a model is only as good as the data behind it. In another project I predicted Bitcoin prices and compared ARIMA, Random Forest and LSTM models. The LSTM worked best, but the more useful lesson was learning to test a time series model honestly, on data it has never seen. The project received Top Project Recognition at our faculty showcase.

After graduating, I worked for a year as a Teaching Assistant at my faculty. Explaining the same idea to many students in different ways made my own understanding much more solid. Before that I did internships in Node.js development at Route and in embedded systems at ITI, and worked for six months as an international account advisor for Vodafone UK at _VOIS, where I learned to communicate clearly in English under pressure. At university I also led teams as Head of Operations at IEEE PUA and Head of HR at GDSC PUA.

The International Master in Data Analytics fits me very well. It is built around exactly what I want to get better at: machine learning, statistics and working with real data. I also like that it is close to the university's machine learning research lab, so the courses are not only theory. Studying in English in an international group suits me, since my whole bachelor's degree was taught and assessed in English.

After the master's, I would like to work as a data scientist or machine learning engineer. I am especially interested in using data in agriculture and health, two areas where I have already seen, in BloomTech and in my Healthcare Database Systems course, how much good data can change real decisions.

I know the program is demanding, and I am ready for that. Thank you for considering my application. I would be very happy to provide any further information.

Sincerely,

Rodayna Emad Mohamed Abdelkader Abdelmawla

Gmail

Rodayna <rodynaemad@gmail.com>

First Project Done!

1 message

WorldQuant University <no-reply@wqu.edu>
To: Rodayna Emad Abdulkader <rodynaemad@gmail.com>

27 January 2023 at 20:52

Dear Rodayna Emad Abdulkader,

Congratulations on completing your first data science project on your path toward gaining these much sought skills.

Think of it, now you can:

- Load data into a pandas DataFrame
- Clean and combine messy data
- Examine the relationship between variables using correlation.
- Create insightful visualizations like histograms, scatter plots, and bar charts.

It's an achievement, and we encourage you to put your new tools and skills to use, building on them as you navigate through your next projects.

Keep up the good work!

All the best,

WQU Team

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WorldQuant University

Gmail

Rodayna <rodynaemad@gmail.com>

Congratulations!

1 message

WorldQuant University <no-reply@wqu.edu>
To: Rodayna Emad Abdulkader <rodynaemad@gmail.com>

2 February 2023 at 02:15

Dear Rodayna Emad Abdulkader,

Congratulations on completing your second data science project on your path toward gaining these much sought skills.

Think of it, now you can:

- Load multiple CSV files into a pandas DataFrame
- Clean messy data with a wrangle function
- Create a machine learning pipeline with feature encoding and imputation
- Create insightful visualizations like a Mapbox scatter plot, a heatmap, and a bar chart for model coefficients

It's an achievement, and we encourage you to put your new tools and skills to use, building on them as you navigate through your next projects.

Keep up the good work!

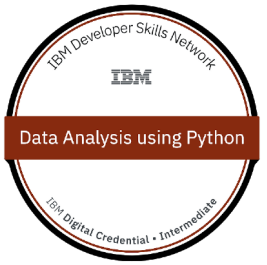
All the best,

WQU Team

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WorldQuant University

In recognition of the commitment to achieve professional excellence



Rodayna Abdulkader

Has successfully satisfied the requirements for:

Data Analysis Using Python

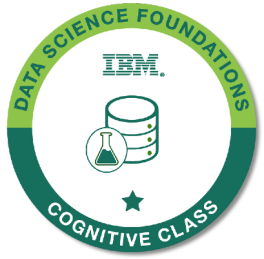
Issued on: 25 NOV 2022

Issued by IBM

Verify: https://www.credly.com/go/LM2BX08S

IBM

In recognition of the commitment to achieve professional excellence



Rodayna Abdulkader

Has successfully satisfied the requirements for:

Data Science Foundations - Level 1

Issued on: 25 NOV 2022

Issued by IBM

Verify: https://www.credly.com/go/6oxqTLEc

IBM

In recognition of the commitment to achieve professional excellence



Rodayna Abdulkader

Has successfully satisfied the requirements for:

Data Science Methodologies

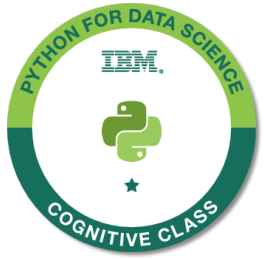
Issued on: 25 NOV 2022

Issued by IBM

Verify: https://www.credly.com/go/YmOdeXvC

IBM

In recognition of the commitment to achieve professional excellence



Rodayna Abdulkader

Has successfully satisfied the requirements for:

Python for Data Science

Issued on: 25 NOV 2022

Issued by IBM

Verify: https://www.credly.com/go/rCta7KNV

IBM



This is to certify that

Rodayna Emad Mohamed

successfully completed and received a passing grade in

Data Science 101

(DS0101EN, provided by **IBM**)

A course on **cognitiveclass.ai**
Powered by **IBM Developer Skills Network.**

Issued by
Cognitive Class

November 25, 2022

Authenticity of this certificate can be validated by going to:
https://courses.cognitiveclass.ai/certificates/75cdf875f9b94580adab602e42b67f10

COGNITIVE CLASS.ai

This is to certify that

Rodayna Emad Abdulkader

successfully completed and received a passing grade in

Data Analysis with Python

(DA0101EN, provided by **IBM**)

A course on **cognitiveclass.ai**
Powered by **IBM Developer Skills Network.**

Issued by
Cognitive Class

November 25, 2022

Authenticity of this certificate can be validated by going to:
https://courses.cognitiveclass.ai/certificates/cb0dc9e8ae7a45c8a773c6f593ad3fa2

COGNITIVE CLASS.ai

This is to certify that

Rodayna Emad Abdulkader

successfully completed and received a passing grade in

SQL and Relational Databases 101

(DB0101EN, provided by **IBM**)

A course on **cognitiveclass.ai**
Powered by **IBM Developer Skills Network.**

Issued by
Cognitive Class

November 25, 2022

Authenticity of this certificate can be validated by going to:
https://courses.cognitiveclass.ai/certificates/6e464a821b5041d2954c40bcfe46aa82

COGNITIVE CLASS.ai

This is to certify that

Rodayna Emad Abdulkader

successfully completed and received a passing grade in

Data Science Methodology

(DS0103EN, provided by **IBM**)

A course on **cognitiveclass.ai**
Powered by **IBM Developer Skills Network.**

Issued by
Cognitive Class

November 25, 2022

Authenticity of this certificate can be validated by going to:
https://courses.cognitiveclass.ai/certificates/4efa4c0268104dbfb3fca940993ccd2e

COGNITIVE CLASS.ai

This is to certify that

Rodayna Emad Abdulkader

successfully completed and received a passing grade in

Python 101 for Data Science

(PY0101EN, provided by **IBM**)

A course on **cognitiveclass.ai**
Powered by **IBM Developer Skills Network.**

Issued by
Cognitive Class

Cognitive Class

November 25, 2022

Authenticity of this certificate can be validated by going to:
https://courses.cognitiveclass.ai/certificates/bff5928a09694f5ba278320cb6cfa1a6

aws training and certification

Certificate of Completion

Rodayna Emad

successfully completed

Machine Learning Essentials for Business and Technical Decision Makers

on

12/26/2022

CERTIFICATE

PROUDLY PRESENTED TO

Rodayna Emad

Fundamentals of a Modern Data Strategy

Sep 30, 2022	*AWS Training and Certification*
Date of Completion	Organizer

aws training and certification

CERTIFICATE

PROUDLY PRESENTED TO

Rodayna Emad

Modern Applications on AWS

Oct 31, 2022

Date of Completion

AWS Training and Certification

Organizer

aws training and certification

LEARNING

Certificate of Completion

Congratulations, Rodayna Emad

Build Your Analytical Skills with Statistical Analysis

Learning Path completed on Aug 25, 2022 at 02:19AM UTC • 1 hour 43 min

By continuing to learn, you have expanded your perspective, sharpened your skills, and made yourself even more in demand.

Top skills covered

Statistical Analysis, Statistics

Head of Content Strategy, Learning

LinkedIn Learning
1000 W Maude Ave
Sunnyvale, CA 94085

Certificate Id: AU51Vn4qGzzmCTRjGbfLQbtY6z15

LinkedIn LEARNING

Certificate of Completion

Congratulations, Rodayna Emad

Data Analytics: Dashboards vs. Data Stories

Course completed on Jul 20, 2022 at 11:26PM UTC • 17 min

By continuing to learn, you have expanded your perspective, sharpened your skills, and made yourself even more in demand.



Head of Content Strategy, Learning

LinkedIn Learning
1000 W Maude Ave
Sunnyvale, CA 94085

Certificate Id: AWj89yBucq_bN3lW_XmagnsrjSa2

LinkedIn LEARNING

Certificate of Completion

Congratulations, Rodayna Emad

SQL for Exploratory Data Analysis Essential Training

Course completed on Jul 23, 2022 at 07:08AM UTC • 44 min

By continuing to learn, you have expanded your perspective, sharpened your skills, and made yourself even more in demand.



Head of Content Strategy, Learning

LinkedIn Learning
1000 W Maude Ave
Sunnyvale, CA 94085

Certificate Id: Acm59jcURxCATIs3b5s-fkL6cod2

STATEMENT OF ACCOMPLISHMENT

#25,086,073

HAS BEEN AWARDED TO

Rodayna Emad

FOR SUCCESSFULLY COMPLETING

Introduction to Python

COMPLETED ON

JUL 19, 2022



Jonathan Cornelissen, CEO

datacamp

STATEMENT OF ACCOMPLISHMENT

#25,178,705

HAS BEEN AWARDED TO

Rodayna Emad

FOR SUCCESSFULLY COMPLETING

Intermediate Python

LENGTH

4 HOURS

COMPLETED ON

SEP 10, 2022

datacamp



Jonathan Cornelissen, CEO

STATEMENT OF ACCOMPLISHMENT

#25,038,778

HAS BEEN AWARDED TO

Rodayna Emad

FOR SUCCESSFULLY COMPLETING

Introduction to SQL

COMPLETED ON

JUL 18, 2022



Jonathan Cornelissen, CEO

datacamp

STATEMENT OF ACCOMPLISHMENT

#25,038,778

HAS BEEN AWARDED TO

Rodayna Emad

FOR SUCCESSFULLY COMPLETING

Intermediate SQL Queries

LENGTH

4 HOURS

COMPLETED ON

JUL 18, 2022



Jonathan Cornelissen, CEO

datacamp

Certificate of Completion

ITI Platform Certifies that

Rodayna Emad

has successfully Completed

Database Fundamentals

- Course time: **2 Hour, 47 Minutes**
- Completion date: **01/09/22**
- Verification code: **FRPYlJmrZB**

Mahara-Tech

Information Technology Institute Platform

Completion certificate issued by the Information Technology Institute platform for each trainee who fulfilled the requirements for passing the course.

مؤسسة مستشفى سرطان الأطفال-مصر
Children's Cancer Hospital Foundation - Egypt
57357

Children's Cancer Hospital Egypt -57357

Department of Continuous Education and Development

Acknowledges that

Rodayna Emad Abdulkader

Has attended "Artificial Intelligence in Healthcare" online course

February 2023

Prof. Manal Zamzam
Director, Continuous Education Department, CCHE-57357

مستشفى سرطان الأطفال 57357 Continuos Education

رحلة علم هدفها الحياه
Knowledge into action

19057 | www.57357.org

Unihance

Certificate

OF COMPLETION

THIS CERTIFICATE IS PRESENTED TO

Rodayna Emad

HAS SUCCESSFULLY COMPLETED THE FOLLOWING COURSE

C++ programming

Thursday, January 26, 2023
DATE

Unihance

Samir ElAdl

Valid Certificate ID

a7c428eb-0e73-46a8-94a9-3d7c23f1a336

Certificate of Achievement



awarded to

Miss Rodayna Emad

Pharos University in Alexandria

The 2022 ICPC ECPC Qualifications Collegiate Programming Contest Day 1
15 - 15 August 2022

11.0

William B. Poucher, Ph. D.
ICPC Executive Director

Certificate of Achievement

The 2022 ICPC ECPC Qualifications Collegiate Programming Contest Day 1

15 - 15 August 2022

11.0

Pharos University in Alexandria

Rodayna Emad

Ahmed Mohamed Ghanem

mostafa khalifa

Saeed Zalat, Coach

William B. Poucher, Ph. D.
ICPC Executive Director

Corporate Social Responsibility

Certificate of Course Completion

Cisco Networking Academy

Entrepreneurship

For completing the Cisco Networking Academy® Entrepreneurship course, and demonstrating the ability to do the following:

- Identify entrepreneurial and transferable job skills that provide value in the marketplace.
- Explain common business and marketing terms.
- Develop a simple business plan.
- Explain the social, legal, and ethical responsibilities of an entrepreneur.
- Analyze business opportunities for a given location and market.
- Explain the role, structure and process for obtaining outsourced services in business.
- Identify, create and expand a business structure.
- Use of social network and media tools to grow business.
- Explain business metrics and financial management.

Rodayna Emad Mohamed

Student

Pharos University in Alexandria (S+Edu)

Academy Name

Egypt

Location

23 Sep 2022

Date

Latifa Elsayes

Instructor

Instructor Signature

Pharos University in Alexandria

Courses Description



Program: Graduate

Faculty: Computer Science&Artificial Intelligence

Degree: BD.in Computer Science&Artificial Int.

Curriculum: Computer Science

Degree Awarded: BD.in Computer Science&Artificial Int.

Graduation Term: (2024-2025) SPRING

اسم الطالب: رودينه عماد محمد عبد القادر عبد المولى

Student Name: Rodayna Emad Mohamed Abdelkader Abdelmawla

Student ID: 202102097

National ID: 30212040200683

Nationality: EGYPT

Date of Birth: 4 Dec 2002

Academic level: Graduated

CGPA: 3.8836

Date: 8/4/2025

COURSE CODE: CS 101

COURSE_DESCRIPTION: Lec 2 & lab 2
Object- class and Methods - Exception handling - Primitive types - Arrays - Data representation in memory - The concept of recursion - Inheritance and polymorphism in OOP

COURSE CODE: CS 106
COURSE_TITLE: Software Engineering
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & lab 2
The changing pressures on software engineering practices: History of the field- definition of software- the software crisis. The Process for Developing Software and its importance The Capability Maturity Model The traditional software engineering process: The lifecycle model- evolutionary software development- incremental software development- spiral model. Prototyping Rapid software development Internet speed web-based application development -End-user development. Agile methods Extreme programming Refactoring

COURSE CODE: DS 102
COURSE_TITLE: Introduction to Data Sciences
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Data Manipulation- Data Analysis with Statistics and Machine Learning- Data Communication with Information Visualization-- Data at Scale -- Working with Big Data

COURSE CODE: MATH 102
COURSE_TITLE: Calculus II
CREDIT_HOURS: 2.000
PREREQUISITE: MATH 101

COURSE_DESCRIPTION: Lec 2 & Tut 2
Definite Integrals – Definite Integrals as Functions – The fundamental Theorem of Calculus - Integration by Parts - Integration by Rationalizing Substitutions - Powers and Products of Trigonometric Functions - Trigonometric Substitutions - Partial Fractions - Evaluating Areas in Cartesian and Polar Coordinates - Evaluating Arc Length in Cartesian and Polar Coordinates - Volumes of Revolutions by the Parallel cross section Method - Volumes of Revolutions by the Shell Method - Mathematical Induction - Roots of Polynomial Equations - Complex Numbers - Determinants and their Properties

COURSE CODE: SS 102
COURSE_TITLE: Introduction to Computer Security
CREDIT_HOURS: 3.000
PREREQUISITE:



CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Newton's Laws and motion. Conservation of energy and momentum. Gravitation Electric charge in matter. Electric forces and fields. DC circuits. Waves and sound

COURSE CODE: UCS 01
COURSE_TITLE: Communication Skills (1)
CREDIT_HOURS: 1.000
PREREQUISITE:

COURSE_DESCRIPTION: Practical 2
To develop students' communication and presentation skills and train them to present themselves and their ideas orally and written in an effective manner that leads them to excellence and leadership in their work and lives.

COURSE CODE: UGB 03
COURSE_TITLE: Computer & Society
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Credit Hours 2
use of personal computers and software. Effects of computer technology on contemporary society. Productivity applications, Creation of Web pages, Online behavior, Internet governance, AI, and cyber security, and societal and ethical issues in computing; privacy, security, censorship, and the changes in work, school, and entertainment fostered by computing. computer impact issues of tomorrow. the meaning of the term "computer ethics" negative impacts that computer technology might bring to individuals.

COURSE CODE: AI 102
COURSE_TITLE: Introduction to Artificial Intelligence
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Reasoning, Intelligent agents, Knowledge representation techniques, Problem solving by searching, Constraint satisfaction problems, Logic programming, Uncertain knowledge and probabilistic reasoning, Planning, Applications

COURSE CODE: CS 104
COURSE_TITLE: Programming II
CREDIT_HOURS: 3.000
PREREQUISITE: CS 103



COURSE_DESCRIPTION: Tutorial 2.5 & Practical 2.5
Students at this level can understand the main points of clear standard input on familiar matters regularly encountered in work, school, leisure, etc. They can deal with most situations likely to arise whilst travelling in an area where the language is spoken. Also, they can produce simple connected text on topics which are familiar or of personal interest, and can describe experiences and events, dreams, hopes & ambitions and briefly give reasons and explanations for opinions and plans.

COURSE CODE: AI 301
COURSE_TITLE: Machine Learning
CREDIT_HOURS: 3.000
PREREQUISITE: AI 201

COURSE_DESCRIPTION: Lec 2 & Lab 2
Feature representation, Similarity, Dimensionality reduction, Supervised learning, Regression, Unsupervised learning, Evaluation, Fundamentals of neural networks (Feedforward and Backpropagation), Reinforcement learning, Applications

COURSE CODE: CS 301
COURSE_TITLE: Computer Networks
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to network architecture, layering, and protocols, Principles of network applications and application layer protocols examples, Socket programming, Introduction to transport layer protocols: Principles of reliable data delivery (error control, congestion control, and flow control), TCP protocol, Introduction to the network layer: Network layer addressing, Routing and forwarding, Principles of routing algorithms

COURSE CODE: CS 305
COURSE_TITLE: Web programming
CREDIT_HOURS: 3.000
PREREQUISITE: CS 103

COURSE_DESCRIPTION: Lec 2 & Lab 2
Web fundamentals, Programming Languages for the Web, HTML Basics, Using HTML, Using CSS and templates, Basics of JavaScript, Programming with JavaScript, Introduction to front end programming, PHP, Django, AngularJS, React, responsive web design, Full stack state management, Security pitfalls and basic solutions. Cloud computing. Cloud Service Providers. Cloud Computing Architecture. Cloud Computing Terminology. Communication-as-a-Service (CaaS). Infrastructure-as-a-Service (IaaS). Monitoring-as-a-Service (MaaS). Software-as-a-Service (SaaS). Platform-as-a-Service (PaaS).

COURSE CODE: CS 307
COURSE_TITLE: Theory of computation
CREDIT_HOURS: 3.000
PREREQUISITE: CS 101



COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to information security, need for security, Legal, ethical, and professional issues in information security, Risk management, Planning for security, Security technologies, and Wireless security technologies, Overview of Cryptography, Security and personnel information security maintenance

COURSE CODE: UGE 01
COURSE_TITLE: English Language (1)
CREDIT_HOURS: 2.000
PREREQUISITE: UGE 00 UGE 00 UGE 00C

COURSE_DESCRIPTION: Tutorial 2.5 & Practical 2.5
Students at this level can understand sentences and frequently used expressions related to areas of most immediate relevance (e.g. very basic personal and family information, shopping, local geography, employment). They can communicate in simple and routine tasks requiring a simple and direct exchange of information on familiar and routine matters, and can describe in simple terms aspects of his/her background, immediate environment and matters in areas of immediate need.Moreover, They can understand the main points of clear standard input on familiar matters regularly encountered in work, school, leisure, etc., and can deal with most situations likely to arise whilst travelling in an area where the language is spoken

COURSE CODE: AI 201
COURSE_TITLE: Intelligent Systems
CREDIT_HOURS: 3.000
PREREQUISITE: AI 102

COURSE_DESCRIPTION: Lec 2 & Lab 2
Definitions of intelligent systems. Optimality vs. speed tradeoff. Problem spaces. Brute-force search (DFS- BFS- uniform cost search). Heuristic search (best-first- A*- IDA*). Local search (hill-climbing- simulated annealing- genetic search). Game-playing methods (minimax search- alpha-beta pruning). Constraint satisfaction (backtracking and heuristic repair). Representation of space and time. Representations of events and actions Probabilistic reasoning. Bayes theorem. Predicate calculus and resolution. Logic programming and theorem proving.AI planning systems. Unsupervised vs. supervised learning. Inductive vs. deductive. Classification vs. clustering vs. prediction. Decision tree learning and neural network learning as examples. Design and development of knowledge-based systems. Knowledge representation mechanisms. Reasoning with uncertainty (non-monotonic logics- certainty factors- fuzzy logic). Knowledge acquisition techniques. Knowledge engineering. Tools for knowledge-based system development

COURSE CODE: CS 201
COURSE_TITLE: Data Structures and Algorithms
CREDIT_HOURS: 3.000
PREREQUISITE: CS 103

COURSE_DESCRIPTION: Lec 2 & Lab 2
Specification and representation of the elementary data types(integers- real- character- logical) - Specification- representation and implementation of basic data structure: Linear data structures (Arrays- lists- Queues- Stacks- Double ended queues- Files) - Non-linear Data Structures (Trees- Binary Trees- Graphs) - Implementation of Linked and dynamic data structures - File layout - Directories: contents and structure - Naming- searching- access- backups - Fundamental file concepts (organization- blocking- buffering) - Sequential files - Non-sequential files

COURSE_TITLE: Discrete Structures
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: lecture 2 & Tutorial 3
Functions- surjections- injections- inverses- composition - Relations: reflexivity- symmetry- transitivity- equivalence relations - Pigeonhole principle - Cardinality and countability - Propositional logic - Logical connectives - Truth tables - Validity - Implication- converse- inverse- negation- contradiction - Predicate logic - Universal and existential quantification - Modus ponens and modus tallness - Formal proofs - Direct proofs - Proof by counterexample - Proof by contraposition - Proof by contradiction - Well orderings - Counting arguments - Permutations and combinations - Solving recurrence relations - Trees - Undirected graphs - Directed graphs - Spanning trees

COURSE CODE: CS 103
COURSE_TITLE: Programming I
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Problem- solving strategies - Debugging strategies - Structured decomposition - concept and properties of algorithms - The role of algorithms in the problem-solving process - Basic syntax and semantics of a higher-level language - Variables, types- and assignment - Conditional and iterative control structures

COURSE CODE: DS 101
COURSE_TITLE: Fundamental of Information Systems
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Tut 2
Introduction to information systems by examining what is an IS, what are the different types, and how businesses use them. In addition, the module will examine the basic technologies, which provide the technical foundation for information systems. The workshops, which run parallel to the lectures, will be used to reinforce an understanding of areas covered in the lectures

COURSE CODE: MATH 101
COURSE_TITLE: Calculus I
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Tut 2
Functions and models, Limits and derivatives, Differentiation rules, Applications of differentiation, Integrals and applications of integration, Techniques of integration, Differential equations, Partial differential equations

COURSE CODE: PHY 101
COURSE_TITLE: Physics I



COURSE CODE: CS 401
COURSE_TITLE: Mobile Application
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Android platform and the Android Studio IDE. Android components- Activities and their lifecycle- Intents UI Design: Widgets and Layouts- UI Events- Event Listeners 2D graphics: Canvas/Drawing using a view Multimedia: Audio playback and Media Player Network support

COURSE CODE: MATH 202
COURSE_TITLE: Linear Algebra
CREDIT_HOURS: 2.000
PREREQUISITE: MATH 102

COURSE_DESCRIPTION: Lec 2 & Tut 2
Matrix theory and linear algebra. Systems of equations- vector spaces- determinants- eigenvalues- similarity- positive definite matrices. Linear operators on finite dimensional vector spaces. The algebraic co-ordinate free methods throughout the course. The scalar field will be arbitrary (even a finite field)- however- in the treatment of vector spaces with inner products- special attention will be given to the real and complex cases. Determinants via the theory of multilinear forms. Variety of examples of the important concepts

COURSE CODE: UEB 02
COURSE_TITLE: Critical Thinking
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Credit Hours 2
Fundamental concepts of critical thinking (logic, argument analysis, rhetoric, reasoning with probabilities, the importance of background knowledge, etc.). The role that critical thinking principles play in good essay writing. how our tribal psychology and political polarization affect our ability to think critically. The importance of critical thinking for personal development, participation in democratic society, and the pursuit of wisdom. how cognitive biases make us prone to errors in how we form beliefs and make judgments. how debates in science often turn on the meanings of key terms, like "theory", "fact", "hypothesis" and "law".

COURSE CODE: UGE 03
COURSE_TITLE: English Language (3)
CREDIT_HOURS: 2.000
PREREQUISITE: UGE 02 UGE 02



COURSE CODE: UGE 02
COURSE_TITLE: English Language (2)
CREDIT_HOURS: 2.000
PREREQUISITE: UGE 01 UGE 01

COURSE_DESCRIPTION: Tutorial 2.5 & Practical 2.5
Students at this level can understand the main points of clear standard input on familiar matters regularly encountered in work school, leisure, etc. They can deal with most situations likely to arise whilst travelling in an area where the language is spoken, produce simple connected text on topics which are familiar or of personal interest. And they can describe experiences and events, dreams, hopes & ambitions and briefly give reasons and explanations for opinions and plans.

COURSE CODE: CS 204
COURSE_TITLE: Computer Architecture
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Fundamentals of processor, memory, and network design, advanced processor and memory design. Example architecture from industry.

COURSE CODE: CS 205
COURSE_TITLE: Algorithms Design
CREDIT_HOURS: 3.000
PREREQUISITE: CS 201

COURSE_DESCRIPTION: Lec 2 & Lab 2
Asymptotic notations, solving recursive relations, Basic analysis measures: Worst and average-case complexity bounds, Amortized analysis, Randomization, Fundamental design strategies: Divide-and-conquer, Dynamic programming, and Greedy methods, String algorithms, Geometric algorithms, Number-theoretic algorithms, Complexity classes, NP complete problems, Approximation algorithms

COURSE CODE: CS 206
COURSE_TITLE: Operating System
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Structuring methods and the layered model - Evolution of hardware/software techniques - Device organization - Interrupts - User/system state and protection - State diagrams - Structures (ready list- process control blocks- and so forth) - Dispatching and context switching - Concurrent execution - Mutual exclusion Deadlock and starvation - Models and mechanisms (semaphores-monitors- rendezvous) - Producer-consumer problems - Preemptive and non-preemptive scheduling - Schedulers and policies - Processes and threads - Deadlines and real-time issues - Overlays- swapping- and partitions - Paging and segmentation - Memory mapped files - Placement and replacement policies - Serial or parallel device - Servers and interrupts - File layout - Directories: contents and structure - Sequential files - Non sequential files - Process and task scheduling - Memory and disk management - Real-time systems



COURSE CODE: CS 202
COURSE_TITLE: Database Systems
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2 & tut 2
Entity-Relationship (E/R) Models - E/R Model Designs and Constraints - Object Definition Language (ODL) - The Relational Model - Converting E/R Models to Relational Designs - Converting ODL Models to Relational Designs - Design of Relational Database Schemas - 3NF and 4NF - SQL and Relational Algebra - Data Definition using SQL - Constraints in Relational Algebra and SQL - Bags vs - Sets- Extended Operators - Extended Operators and Views

COURSE CODE: CS 203
COURSE_TITLE: Logic Circuit Design
CREDIT_HOURS: 3.000
PREREQUISITE: PHY 101

COURSE_DESCRIPTION: Lec 2 & Lab 2
Provides a modern introduction to logic design and the basic building blocks used in digital systems, in particular digital computers. It starts with a discussion of combinational logic: logic gates, minimization techniques, arithmetic circuits, and modern logic devices such as field programmable logic gates. The second part of the course deals with sequential circuits: flip-flops, synthesis of sequential circuits, and case studies, including counters, registers, and random-access memories. State machines in complex systems using programmable logic devices. Different representations including truth table, logic gate, timing diagram, switch representation, and state diagram.

COURSE CODE: MATH 201
COURSE_TITLE: Probability and Statistics
CREDIT_HOURS: 2.000
PREREQUISITE: MATH 101

COURSE_DESCRIPTION: Lec 2 & Tut 2
Elementary introduction to probability and statistics with applications. Topics include basic combinatory- random variables- probability distributions- Bayesian inference- hypothesis testing- confidence intervals- linear regression

COURSE CODE: UEB 01
COURSE_TITLE: Entrepreneurship & Innovation
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Credit Hours 2
Definitions and evolution of entrepreneurship and innovation. Role of innovation in entrepreneurship and organizational development. Innovation as a capability/ core competence. Understanding entrepreneurial and innovative individuals. Role of innovation in new growth firm creation (independent entrepreneurship). Role of innovation in established enterprise development (corporate entrepreneurship). Role of innovation in high growth or gazelle firms. Role of technology in innovation. Role of champions and culture in innovation. Implementing innovation. Measuring innovation performance. Some key issues in innovation management.



COURSE_DESCRIPTION: Lec 2 & Lab 2
Systems Modeling Concepts- Modeling formalisms and their simulators- Cellular automata- Discrete event systems modeling- Continuous system modeling- Bond Graph approach to modeling physical systems- Stochastic systems- Hybrid (continuous + discrete) systems - Abstraction models- Distributed systems- verification and validation mechanisms

COURSE CODE: CS 209
COURSE_TITLE: Advanced Programming
CREDIT_HOURS: 3.000
PREREQUISITE: CS 104

COURSE_DESCRIPTION: Lec 2 & Lab 2
Object-oriented application programming. Applications such as standalone, web, enterprise solutions. Most in-demand programming languages, a standard for enterprise software, web-based content, games and mobile apps, Android operating system. Multiple software platforms, program written on Mac OS X, for example, could also run in a Windows environment.

COURSE CODE: CS 304
COURSE_TITLE: System Programming
CREDIT_HOURS: 3.000
PREREQUISITE: CS 206

COURSE_DESCRIPTION: Lec 2 & Lab 2
System calls and how they interact with the kernel. File manipulation looking at how to create, delete, and edit files using system call. Memory management and how exactly Unix deals with virtual memory addressing. Processes: Threads. Synchronization issues that arise when using processes and threading

COURSE CODE: CS 308
COURSE_TITLE: Computer Graphics
CREDIT_HOURS: 3.000
PREREQUISITE: MATH 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Raster and vector graphics system - Video display devices - Physical and logical input devices - Issues facing the developer of graphical systems - Hierarchy of graphics software - User interface - Half tones - Font generation: outline vs - bitmap - Representation of polyhedral objects - Scan conversion of 2D primitive- forward differencing - Tessellation of curved surfaces - Homogeneous coordinates - Affine transformations (scaling- rotation- translation) - Viewing transformation. Clipping. Hidden surface removal methods. Z-buffer and frame buffer- color channels (a channel for opacity). Color models (RGB- HIVS- CYM). Light source properties - material properties - ambient- diffuse- specular reflections. Reflection model. Rendering of a polygonal surface- flat shading- Texture mapping- bump texture- environment map. Ray tracing. Image synthesis- sampling techniques- Parametric polynomial curves and surfaces. Implicit curves and surfaces. Bezier curves and surfaces- control points- B-spline curves and surfaces- local editing- control points. NURBS curves and surfaces. Constructive Solid Geometry for solid modeling. Boundary Representation of solids (B-Rep).

COURSE CODE: SS 302
COURSE_TITLE: Internet of Things

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COURSE_DESCRIPTION: Lec 2 & Tut 3
Deterministic Finite Automata (DFA) and Regular Languages - Non-deterministic Finite Automata (NFA) - Equivalence of NFAs and DFAs - Closure Properties - Regular Expressions - The Pumping Lemma - Pushdown Automata - Pushdown Automata - Context Free Languages - Ambiguity and Chomsky Normal Form - Turing Machines - Turing Machines Variants - Enumerators - Decidability - Diagonalization - The Halting Problem - Reducibility - Time Complexity - P and NP - Poly-time reductions - NP-Completeness

COURSE CODE: MATH 301
COURSE_TITLE: Numerical Computations
CREDIT_HOURS: 2.000
PREREQUISITE: MATH 202

COURSE_DESCRIPTION: Lec 2 & Tut 2
Review of Mat lab Programming Language - Additional DATA Types – Derived DATA Types - Debugging Mat lab Programs – Procedures Applications: Interpolation - Numerical Integration and differentiation - Solution of Non-linear Equations - Solution of Systems of Linear Equations - Initial Value Problems - Boundary Value Problems - Elements of Grid Methods for Partial Differential Equations

COURSE CODE: SS 402
COURSE_TITLE: Block chain computing
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to block chains and cryptocurrencies, Distributed ledger technologies, Trust and vulnerabilities, Consensus mechanisms, Basics of contract law, Smart contracts, Regulations and laws of block chains, Security of block chains

COURSE CODE: AI 302
COURSE_TITLE: Robotics
CREDIT_HOURS: 3.000
PREREQUISITE: AI 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Foundations and principles of robotic kinematics, Transformations, Forward kinematics, Inverse kinematics, Differential kinematics (Jacobians), Manipulability and basic equations of motion, Programming on robot arms, Applications, localization and mapping, Kalman filters

COURSE CODE: CS 207
COURSE_TITLE: System simulation and modelling
CREDIT_HOURS: 3.000
PREREQUISITE: CS 101



CREDIT_HOURS: 3.000
PREREQUISITE: CS 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Fundamentals of the Internet of Things (IoT), Impact of IoT on society, Business process in IoT, the design and implementation of IoT devices as embedded systems, Hardware and software components for IoT, IoT node authentication, Developing smart cities, IoT security

COURSE CODE: CS 310
COURSE_TITLE: Human Computing Interaction
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Mapping - Affordances - Constraints - Seven Stages of Action - Schneiderman's 8 Golden Rules - Information Visualization - Model Human Processor - Keystroke Level Model - Fit's law - Input devices (Keyboard- Pointing- Voice) - Output devices (Displays- Color- Sound) - Interaction Styles (direct manipulation- menu selection- form-fill-in- command languages) - Windows - Icons - Menus - Dialogue Boxes - Concepts (grids- simplicity- consistency- white space) - Context Sensitive Help - Tutorials - Reference Material - Cognitive Walkthrough - Heuristic Evaluation - Expert Reviews - Controlled Experiments (subjects- dependent & independent variables- statistics) - Synchronous / Asynchronous tools - Audio / Video - Shared Workspaces

COURSE CODE: CS 403
COURSE_TITLE: Advanced computer networks
CREDIT_HOURS: 3.000
PREREQUISITE: CS 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Principles, architectures, and protocols used in modern networked systems, such as the Internet, wireless and mobile networks, high performance networks and data center networks. Existing technology in building large, complex networked systems, and provide concrete experience of the challenges through a series of lab exercises. Wireless networks and mobility issues at the network and transport layer (Mobile IP and micro mobility protocols, TCP in wireless environments). Data center and high-performance networking. Network virtualization.

COURSE CODE: CS 405
COURSE_TITLE: Distributed Systems and Cloud Computing
CREDIT_HOURS: 3.000
PREREQUISITE: CS 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to Cloud Computing, Cloud computing platforms and infrastructure, Parallel programming in the Cloud, Distributed storage systems, Virtualization, Cloud security, Cloud performance

COURSE CODE: CS 491
COURSE_TITLE: Project I

COURSE_DESCRIPTION: Lec 2 & Lab
Success, Reflections, Terms and Concepts. Competencies, Standards, Software Lifecycle Models, Software Quality Assurance. Software Project Management, Adoption of PM Practices. Reviews and Inspections, Software Testing, Testing Effort, Advances in Testing. Software Quality as a Profession. Metrics and Analysis, Software Configuration Management Release Management.

COURSE CODE: CS 404
COURSE_TITLE: Compiler and Language Theory
CREDIT_HOURS: 3.000
PREREQUISITE: CS 304

COURSE_DESCRIPTION: Lec 2 & Lab 2
Principles and practices for design and implementation of compilers. Specific topics include lexical analysis; parsing theory; symbol tables; type systems; scope; semantic analysis; intermediate representations; runtime environments; code generation; and basic program analysis and optimization. Students construct a compiler for a simple object-oriented language during course programming projects

COURSE CODE: CS 492
COURSE_TITLE: Project II
CREDIT_HOURS: 3.000
PREREQUISITE: CS 491

COURSE_DESCRIPTION: Lec 2 & Lab 2
Demonstrate the advanced skills they have learned in an applications area of their own choosing in project1. projects must include a research component

COURSE CODE: DS 302
COURSE_TITLE: Data Science Methodology
CREDIT_HOURS: 3.000
PREREQUISITE: DS 102

COURSE_DESCRIPTION: Lec 2 & Lab 2
The data science framework, Business understanding, RFM modelling, Data collection and preparation, Structured and unstructured data, Cleaning and Exploration, Data visualization, Market basket analysis, Modeling, Evaluation, Feedback, Use cases

COURSE CODE: DS 307
COURSE_TITLE: Bioinformatics Programming and Scripting
CREDIT_HOURS: 3.000
PREREQUISITE: CS 201



CREDIT_HOURS: 3.000
PREREQUISITE: CS 304

COURSE_DESCRIPTION: Lec 2 & Lab 2
Demonstrate the skills they have learned in an applications area of their own choosing. projects must include a research component

COURSE CODE: DS 411
COURSE_TITLE: Healthcare Database systems
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
The course aims to introduce the principles of health care organization and policy in a comparative perspective nationally and internationally, identify the key characteristics and components of health care systems, assess each health care system's strengths and weaknesses, and discuss the recent health care reform efforts

COURSE CODE: UCS 02
COURSE_TITLE: Communication Skills (2)
CREDIT_HOURS: 1.000
PREREQUISITE: UCS 01

COURSE_DESCRIPTION: Practical 2
To provide a professionally-oriented, skills-based learning that helps the students to get appropriate jobs and emphasizes on providing the community with qualified graduates show excellence in their career.

COURSE CODE: CS 309
COURSE_TITLE: Embedded Systems
CREDIT_HOURS: 3.000
PREREQUISITE: CS 201

COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to computer architecture, Sensors, Actuators, Input-Output (I/O) modeling, I/O Interfacing, Microcontrollers, Real-time operating systems, Energy-aware design, Case studies and applications of embedded control systems, Recent trends and developments

COURSE CODE: CS 311
COURSE_TITLE: Software Quality Assurance and Testing
CREDIT_HOURS: 3.000
PREREQUISITE: CS 106



COURSE_DESCRIPTION: Lec 2 & Lab 2
Algorithms for solving various biological problems with a handful of programming challenges helping implement these algorithms in Python. offers a gently paced introduction to Bioinformatics Specialization





